

OMAR NASSER MAHMOUD

AI ENGINEER

+201285407011 • omarr.elhawyy@gmail.com • linkedin.com/in/omar-nasser-aab951303 • github.com/OmarElHawy
Alexandria, Egypt

SUMMARY

AI & Data Science undergraduate with hands-on experience in machine learning, data analysis, and LLM-based applications. Completed multiple internships applying data preprocessing, exploratory data analysis, prompt engineering, and Retrieval-Augmented Generation (RAG) to real-world AI systems. Proficient in Python, modern AI/ML frameworks, and n8n automation, with a focus on building scalable, data-driven AI solutions and intelligent workflows.

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, Pandas, NumPy, PyTorch, TensorFlow, scikit-learn, Matplotlib, Seaborn

AI / ML: Machine Learning, Deep Learning, Generative AI & LLMs, NLP, Computer Vision, RAG, Prompt Engineering

Tools & Platforms: Git, Docker, FAISS, LangChain, Gradio, FastAPI, N8N Automation, APIs

EXPERIENCE

Intern – Digital Egypt Pioneers Initiative (DEPI)

AI Automation Workflow with N8N Professional | 07/2026 – Present

- Designed AI-powered automation workflows using n8n to support productivity and operational efficiency.
- Applied Generative AI, Prompt Engineering, and AI Agent concepts to practical automation use cases.
- Developed AI automation workflows integrating Slack, Web, and Email with intelligent agents.
- Incorporated security, governance, and responsible-AI principles into scalable automation designs.
- Delivered capstone automation projects modeled on real-world operational scenarios.

Intern – Information Technology Institute (ITI)

Advanced AI | 07/2026 – Present

- Applied Python to practical Artificial Intelligence use cases and end-to-end workflows.
- Implemented Computer Vision techniques, including YOLO-based object detection with pretrained models.
- Built and evaluated NLP components for real-world AI solutions.
- Worked hands-on with Large Language Models (LLMs) and Generative AI application design.
- Implemented Retrieval-Augmented Generation (RAG) pipelines for knowledge-based LLM applications.
- Delivered a final capstone AI project integrating concepts covered across the training program.

Intern – Information Technology Institute (ITI)

Building RAG Agents with LLMs | 01/2026 – 02/2026

- Designed and developed Retrieval-Augmented Generation (RAG) pipelines to enhance LLM responses with external knowledge sources.
- Integrated vector databases for semantic search and efficient document retrieval.
- Built end-to-end AI agents combining retrieval, prompt engineering, and response generation.
- Optimized prompt structures to improve accuracy, relevance, and contextual understanding.
- Evaluated model performance across different retrieval and ranking strategies.

Intern – Information Technology Institute (ITI)

Generative AI | 09/2025 – 10/2025

- Built applications using Large Language Models (LLMs) for text generation, summarization, and Q&A tasks.
- Applied prompt engineering techniques to improve model outputs and reduce hallucinations.
- Evaluated different LLM frameworks and APIs for real-world application scenarios.
- Delivered small-scale generative AI applications demonstrating practical use cases.

Intern – National Telecommunication Institute (NTI)

Data Analysis with Python | 09/2025 – 10/2025

- Collected, cleaned, and preprocessed structured and unstructured datasets using Python (Pandas, NumPy).
- Conducted exploratory data analysis (EDA) to identify trends, patterns, and anomalies.
- Visualized insights using Matplotlib and Seaborn to support data-driven decision-making.
- Applied statistical analysis techniques to interpret data and generate actionable insights.

Intern – WE – Telecom Egypt

Artificial Intelligence | 08/2025 – 09/2025

- Completed foundational training in machine learning workflows and the AI model lifecycle.
- Assisted in data preparation, preprocessing, and feature engineering for AI models.
- Contributed to real-world AI use cases by analyzing datasets and preparing training inputs.
- Collaborated with teams to translate business requirements into AI solutions.

PROJECTS

Arabic (Egyptian) Speech to Text

- Fine-tuned Whisper-small (244M params) using PyTorch on 20,000 Mozilla Common Voice Arabic samples, freezing the CNN encoder and 6 transformer encoder blocks, training only the 6 decoder blocks to prevent overfitting on limited data.
- Built the full audio pipeline: 16kHz resampling, log-mel spectrogram (80 freq bins × 3000 time frames), autoregressive Arabic token decoding.
- Applied Arabic-specific text normalization: diacritic (tashkeel) removal, Alef/Teh Marbuta variant unification, Tatweel stripping.
- Trained with mixed-precision (FP16), gradient accumulation (effective batch size 8), and early stopping (patience=3) on a single RTX 3060 6GB GPU in approximately 3.8 hours.
- Achieved 24.24% WER / 11.14% CER; deployed a real-time Gradio demo supporting live microphone input and file upload.

Automated Timetable Generation (CSP)

- Modeled university scheduling as a Constraint Satisfaction Problem across courses, instructors, rooms, and time slots, solved with a custom Greedy Search heuristic (backtracking planned as a future extension).
- Built a centralized constraint validation layer enforcing hard constraints: no instructor/room/section double-booking, room type/capacity compatibility, and full session coverage for multi-part courses.
- Designed the system to be data-driven – regenerates the full schedule automatically from updated Excel/CSV inputs with no code changes.
- Generated 4 distinct interactive HTML views (main timetable, professor schedules, assistant professor schedules, room availability) from a single solved schedule.
- Modeled against a realistic dataset spanning 4 undergraduate levels of a CS & IT department (based on EJUST's structure).

SmartDoc RAG Agent

- Built a production-style Retrieval-Augmented Generation (RAG) assistant for querying large PDF/DOCX documents (contracts, insurance policies, reports) through a conversational interface.
- Designed a modular pipeline: document ingestion (PDF/DOCX chunking), FAISS vector store, LangChain LCEL retrieval chain, and a pattern-based safety guardrail.
- Deployed via Gradio UI and a FastAPI/LangServe REST API, with built-in evaluation metrics (response time, grounding score).
- Built as part of the NVIDIA DLI Course Workshop.

Blood Cell Image Analysis

- Built a two-phase pipeline using scikit-learn for automated white blood cell (WBC) classification from microscope images.
- Phase I: implemented image enhancement techniques from scratch (histogram equalization, Gaussian/median filtering, Butterworth frequency filters).
- Phase II: LAB color-space segmentation to isolate nuclei, fused CNN features (VGG16/19, ResNet50, EfficientNetB0 via TensorFlow/PyTorch) with LBP/GLCM texture descriptors, reduced with PCA, classified with SVM.

- Achieved 98.75% accuracy and 0.9999 AUC-ROC (best config: EfficientNetB0 + SVM).

EDUCATION

Bachelor's Degree

Egypt Japan University of Science and Technology | 09/2023 – Present, Egypt

CERTIFICATIONS

NVIDIA DLI Training Program: Generative AI (Advanced Level)

Information Technology Institute (ITI)

Jan 2026 – Feb 2026 | 100 hrs | Deep Learning Basics, Building RAG Agents with LLMs, Capstone Project

NVIDIA DLI Summer Training Program: Generative AI (Beginner Level)

Information Technology Institute (ITI)

Sep 2025 – Oct 2025 | 35 hrs | Python Fundamentals, Generative AI Explained, Building LLM Applications with Prompt Engineering, Augmenting LLMs with RAG

Python for Data Analysis

National Telecommunication Institute (NTI) – Digital Egypt Youth Program

Sep 2025 – Oct 2025 | 72 hrs | Score: 98%

Artificial Intelligence – Basic Level (Machine & Deep Learning)

WE – Telecom Egypt

Aug 2025 – Sep 2025

STRENGTHS & LANGUAGES

Strengths: Ownership, Self-Learning, Time Management, Collaboration, Technical Communication, Problem Solving

Languages: English (Fluent), Arabic (Native)